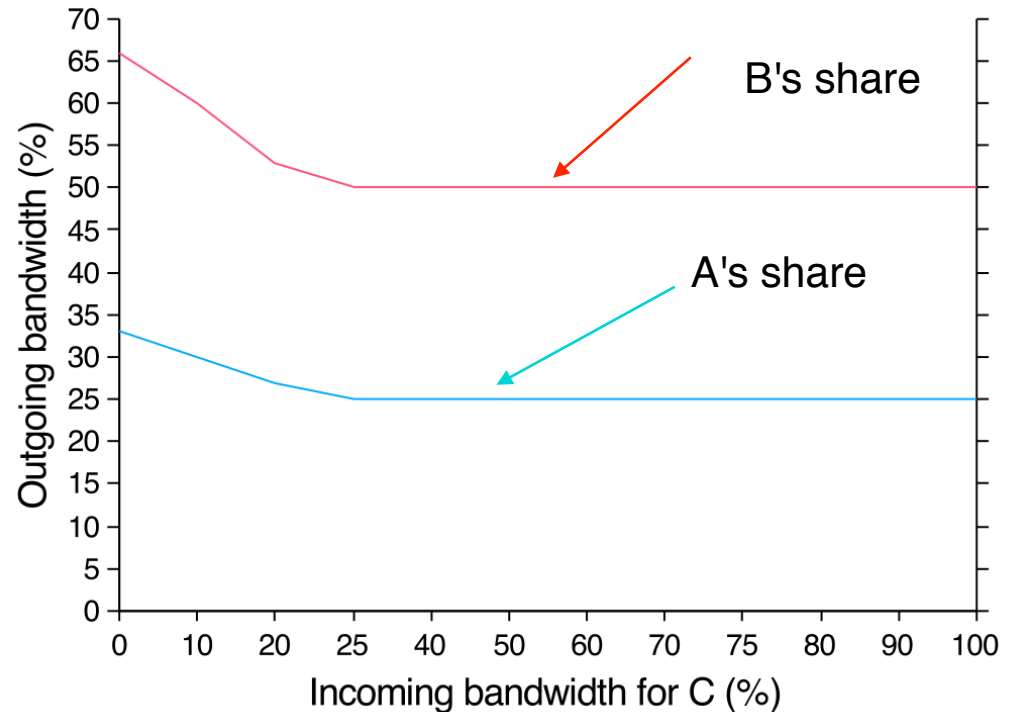
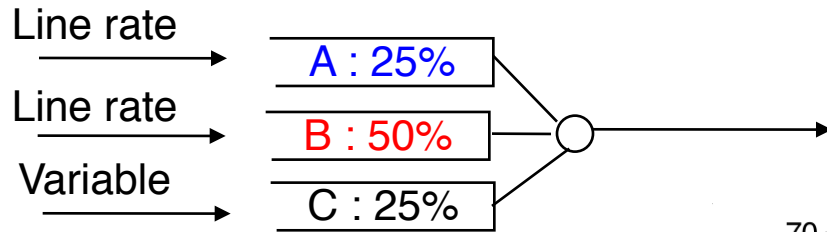


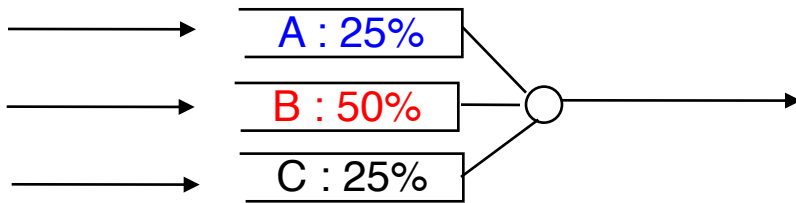
INGI2142

Integrated and Differentiated
services

Weighted Round-Robin



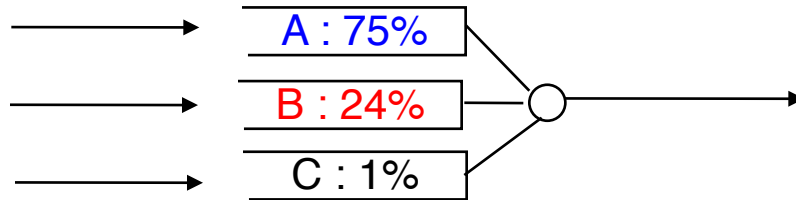
Weighted Round Robin



- Input Traffic pattern
 - A : 20%, regular
 - B: 60%
 - C: 50%

Weighted Round Robin

- What is the impact of a burst of traffic ?

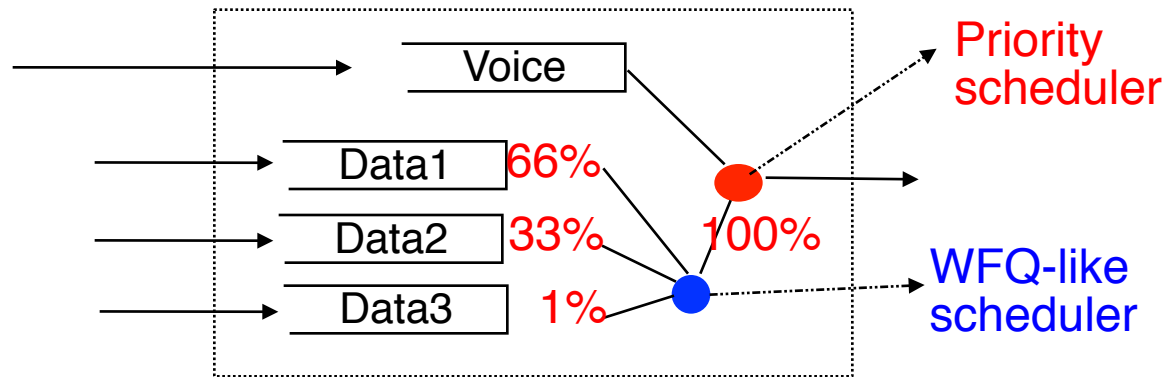


- Input Traffic pattern
 - A : 20%, regular
 - B: 24%
 - C: 100%

Schedulers and buffer acceptance

- Which buffer acceptance strategy should be used with DRR or WDRR ?
 - Can we identify the flow/queue that is causing congestion ?
 - Which packet should be discarded ?

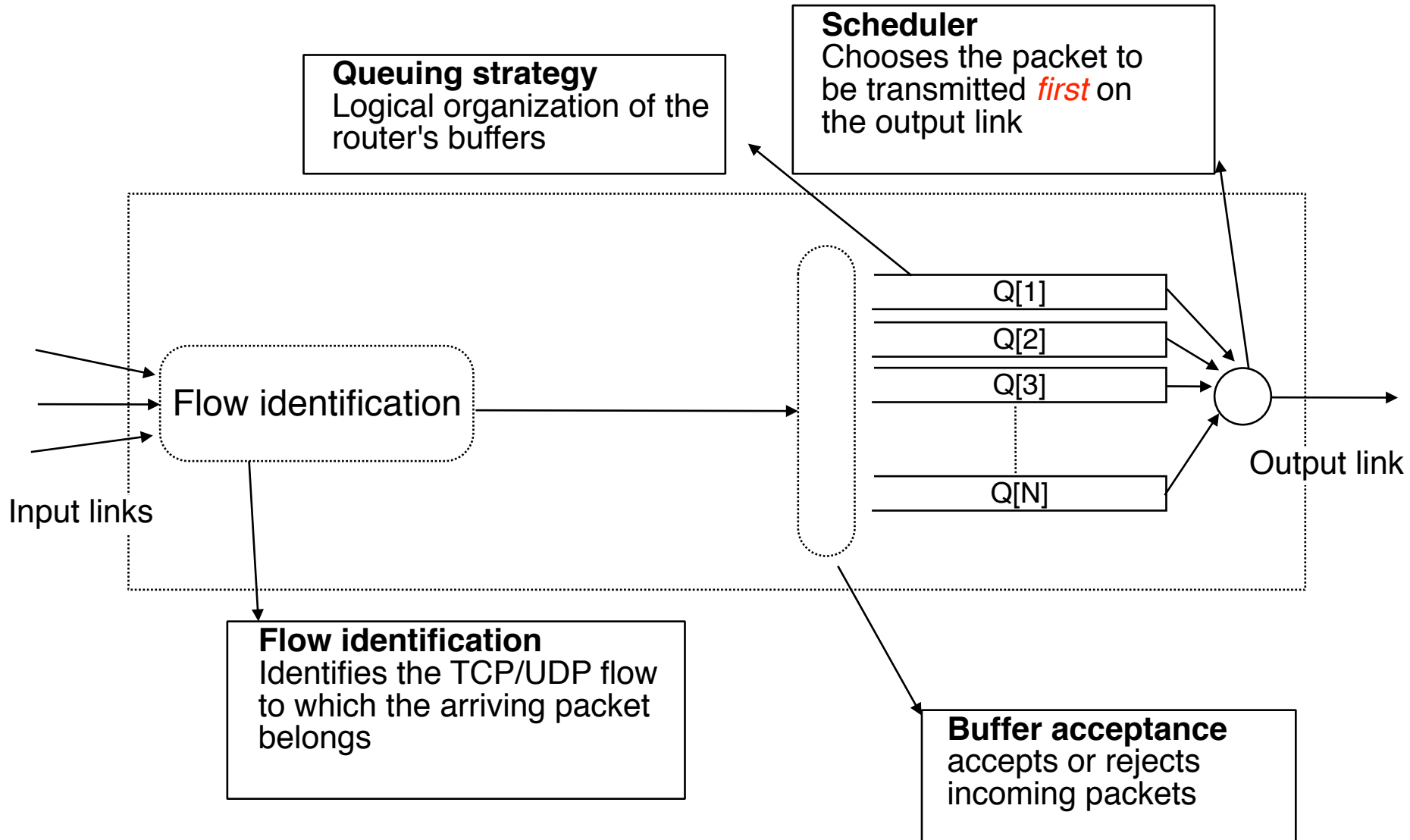
Combining schedulers



- Input Traffic pattern
 - 10% voice, regular
 - 30% data1
 - 60% data2
 - 50% data3

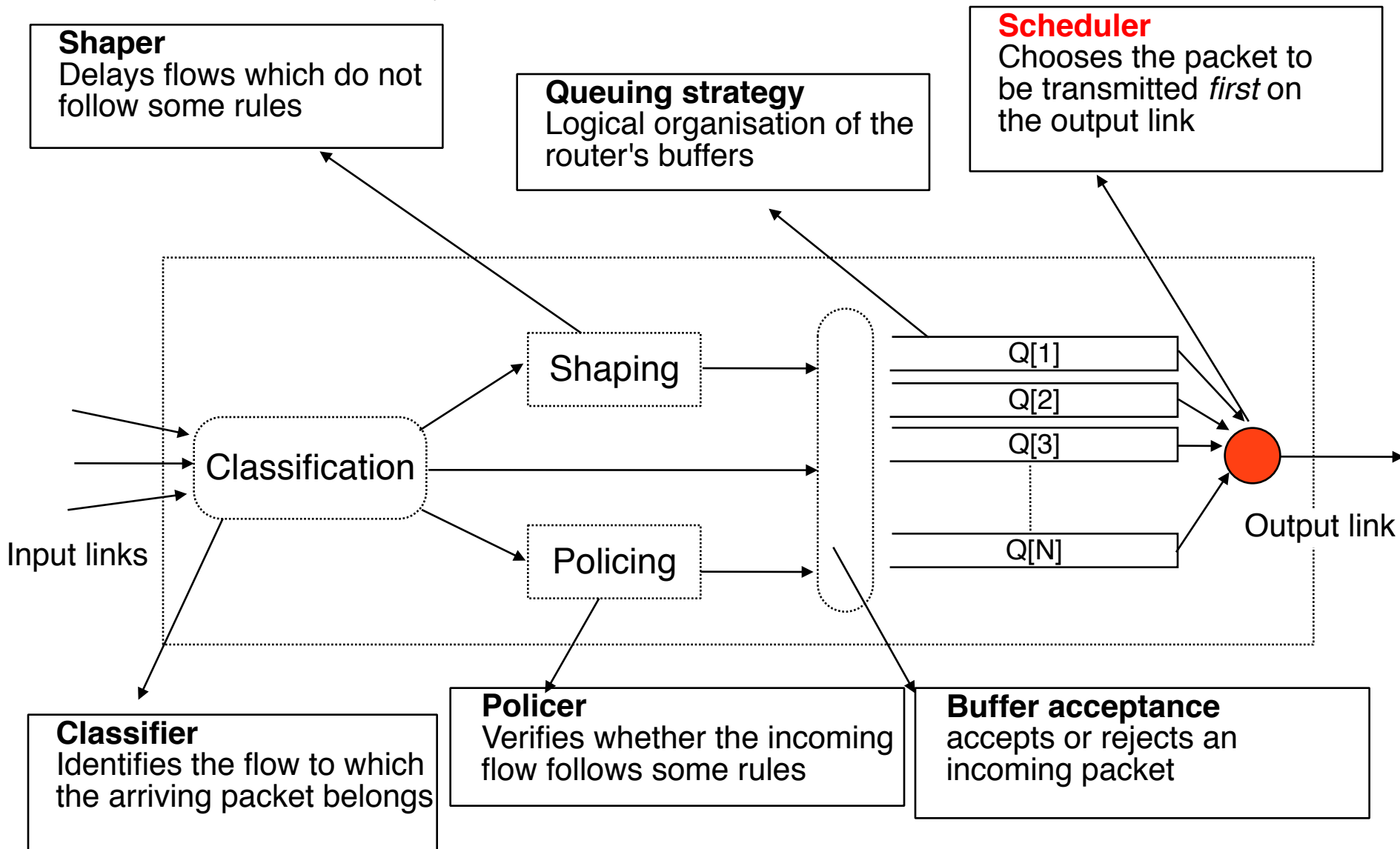
The traffic control building blocks

best effort router

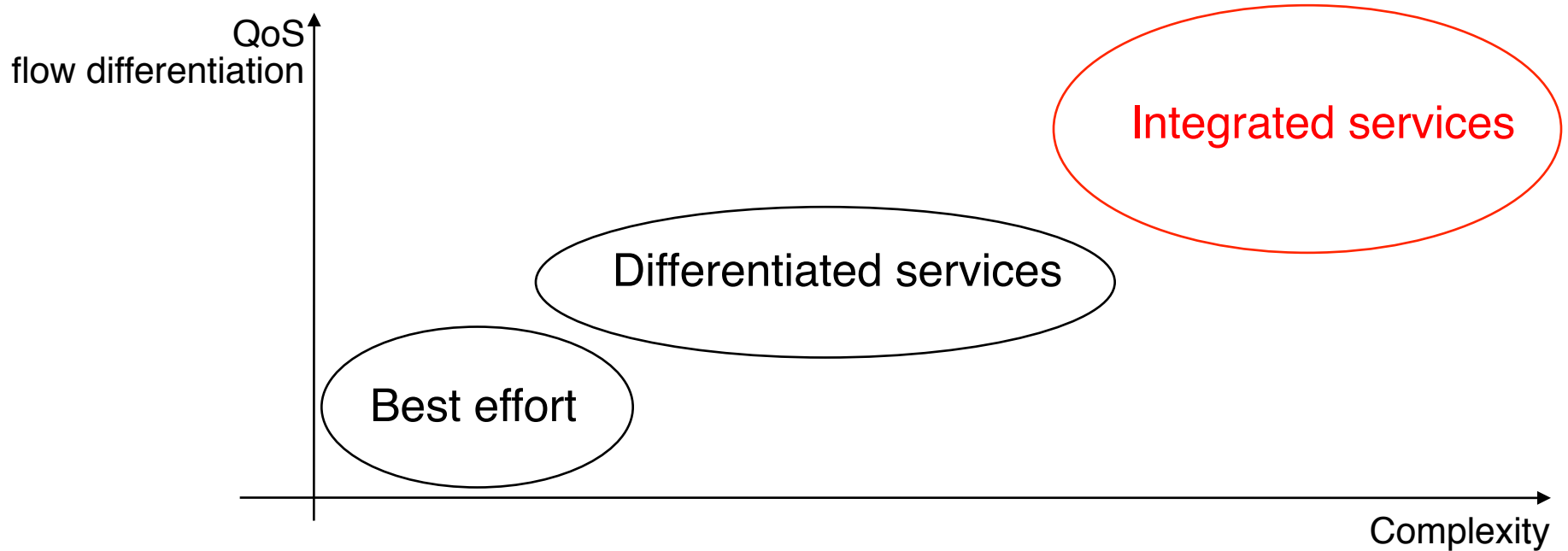


The traffic control building blocks

QoS-enabled router



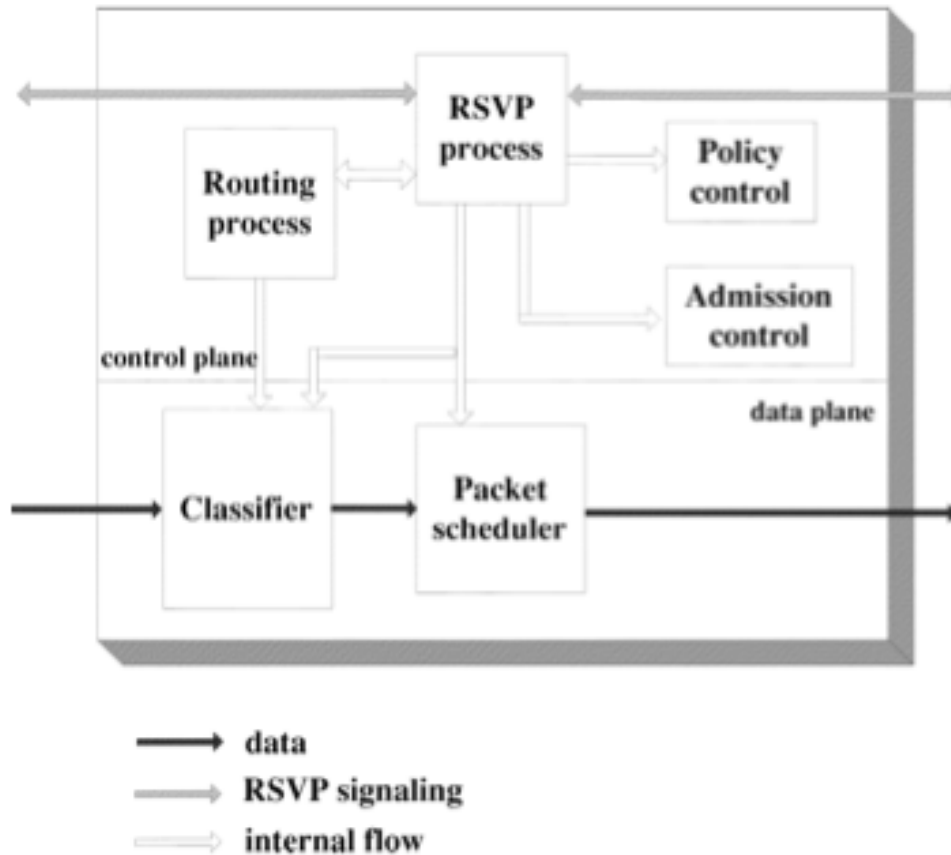
How to build services from these building blocks ?



Integrated services

- What are the key principles of the integrated services model ?
 - What is the architecture of an Integrated Services router ?
 - How to use RSVP for end-to-end flows ?
 - What are the traffic control building blocks used by Integrated services

Architecture of an integrated services router



What is an integrated services flow ?

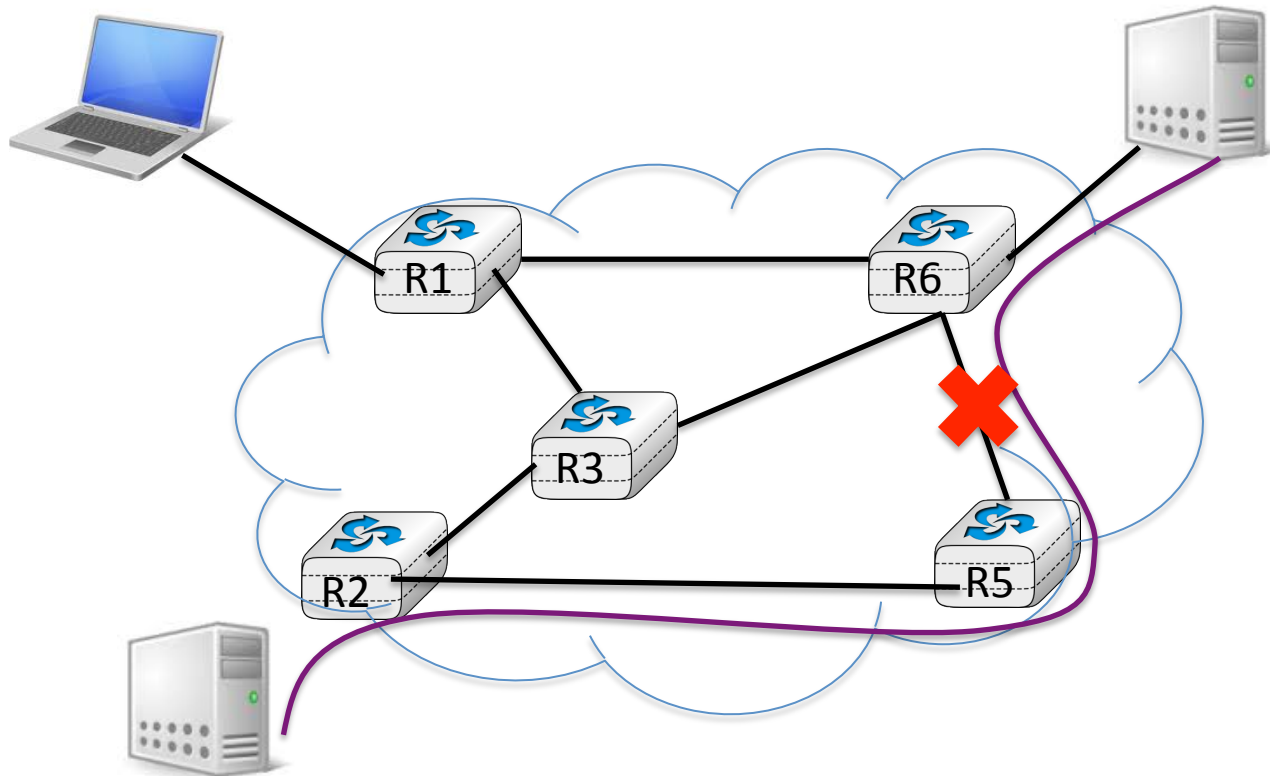
- How do routers identify flows that benefit from integrated services guarantees ?
- RSVP flows are unidirectional layer 4 flows
 - RSVP also supports multicast but we ignore this part of RSVP currently

The Resource Reservation Protocol

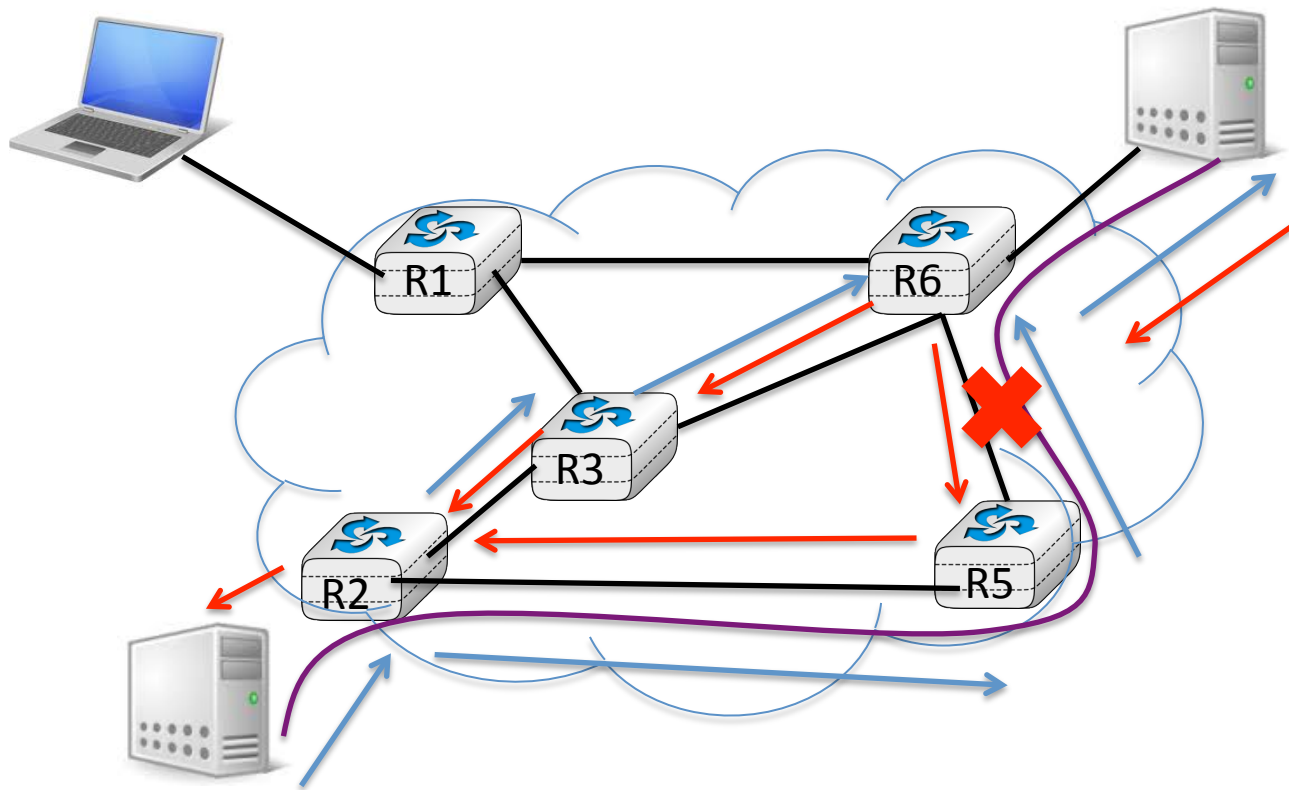
- What are the key principles of this protocol ?
 - What is the role of the PATH message ?
 - What is the role of the RESV message ?
 - How does it interact with IP routing protocols ?

How to deal with link failures ?

- What happens to reservations when a link fails on a path ?

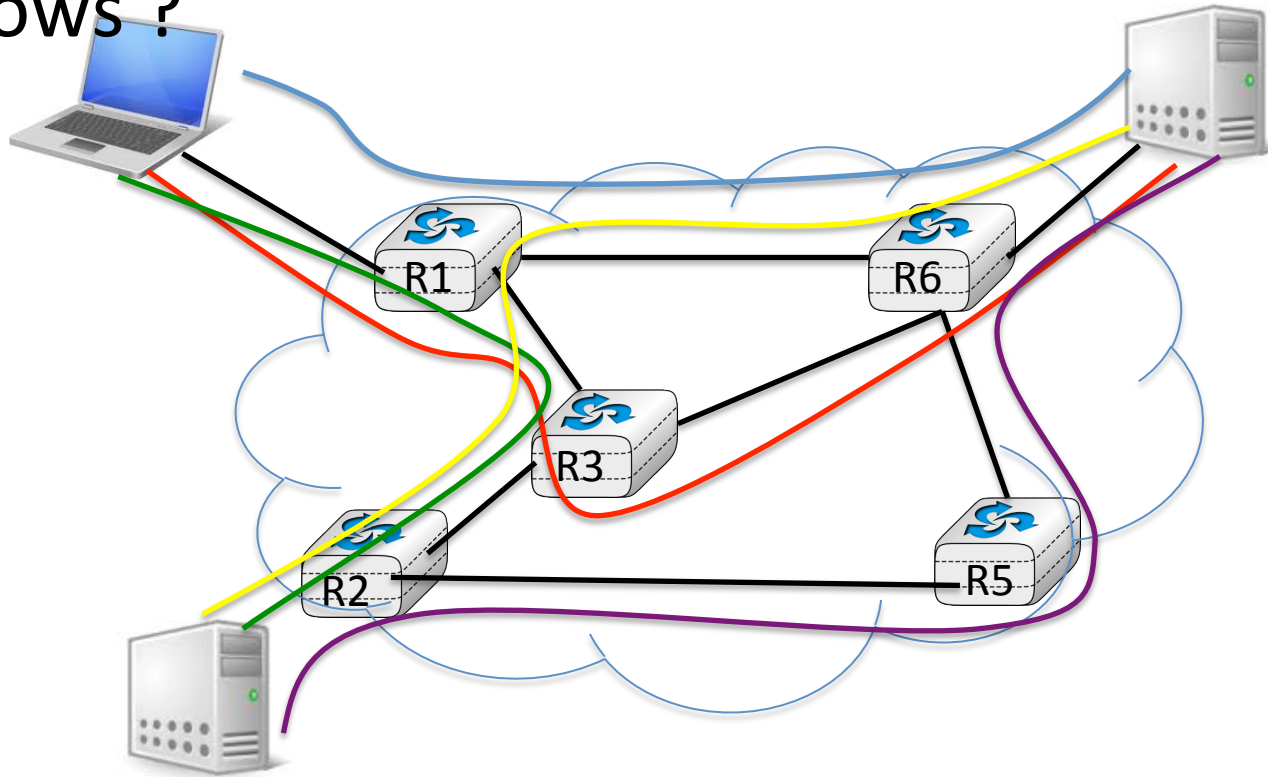


RSVP's soft state



The Resource Reservation Protocol

- Which paths can be used for integrated flows ?



Integrated services

- Three standardised services have been defined
 - The Guaranteed Service
 - The Controlled Load service
 - The Null service

The Guaranteed Service

- For applications requiring strict bounds on the end-to-end delay
 - traffic source is policed by token bucket (r,b)
 - Routers must ensure that each flow has bandwidth $R > r$
 - and guarantee a bound on the maximum packet delay through the router
 - some schedulers can provide this bound

The Controlled Load service

- Provides a similar service as an unloaded network
 - Each flow has a bandwidth guarantee but no delay guarantee
 - How would you provide this service by using schedulers inside the routers ?

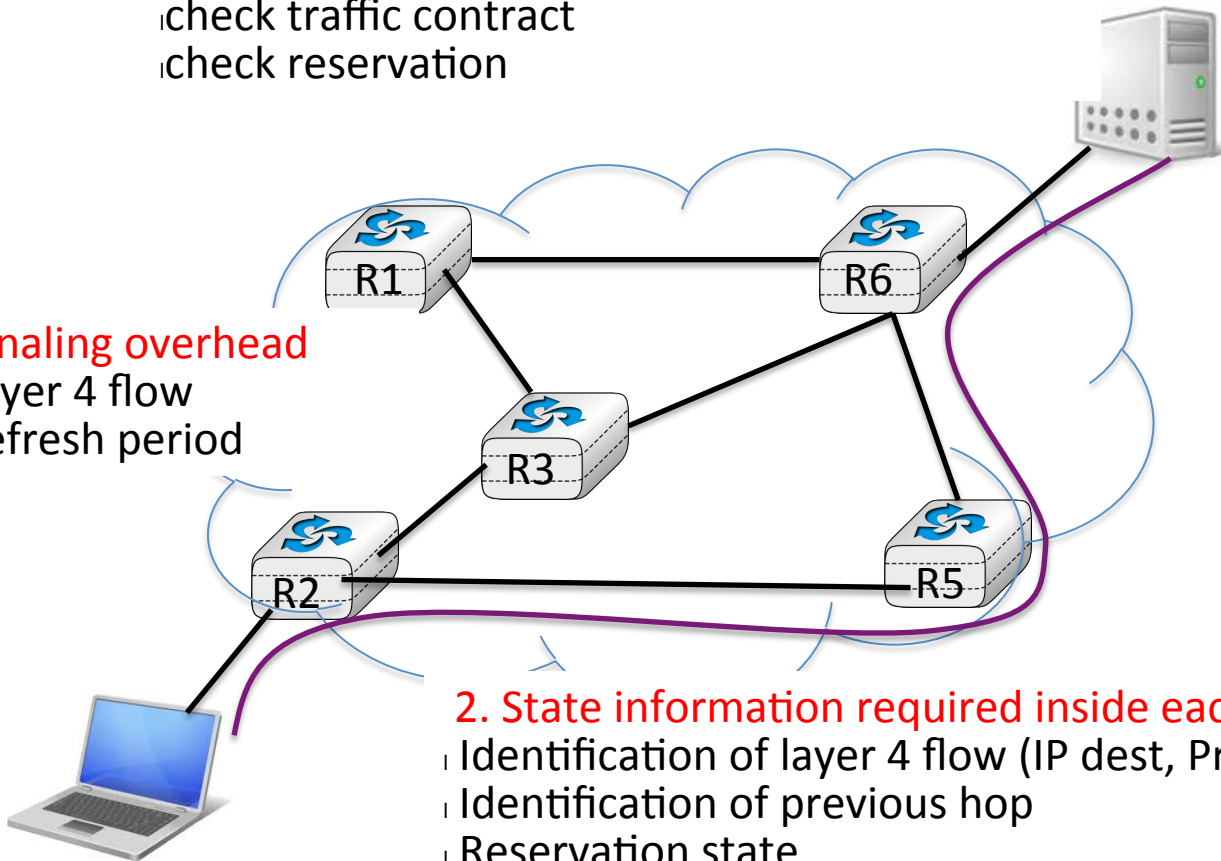
Issues with integrated services

3. Per data packet processing

- | find layer 4 flow state
- | check traffic contract
- | check reservation

1. signaling overhead

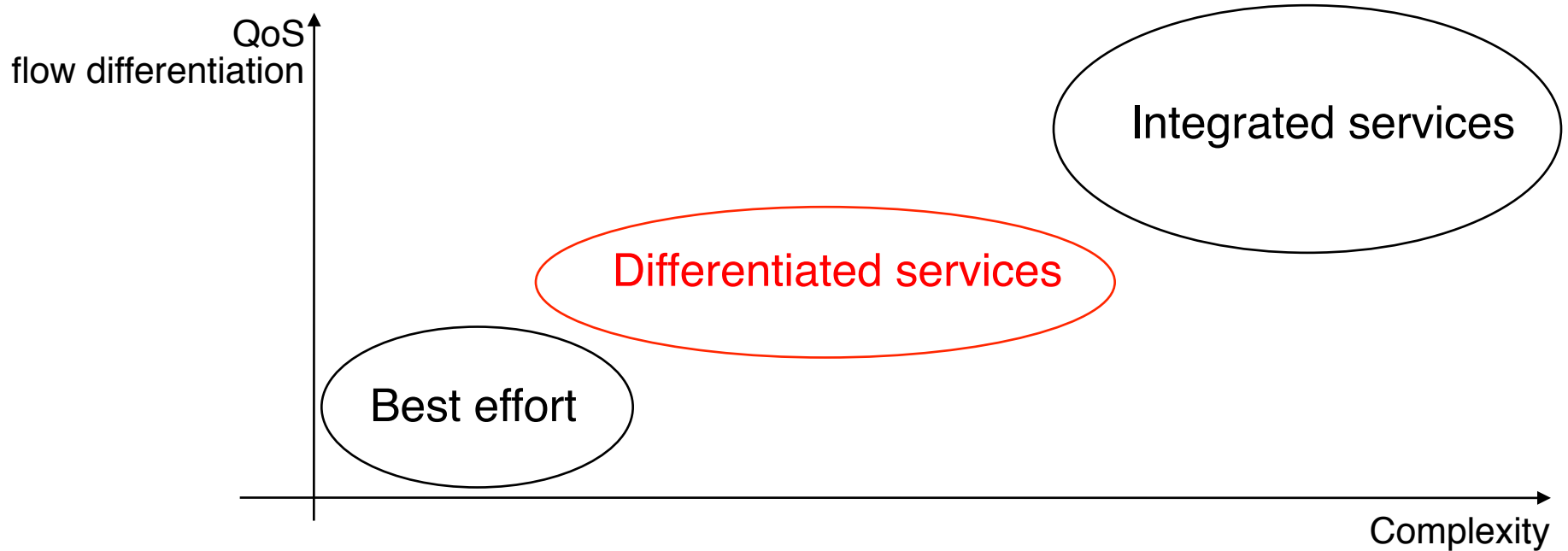
- | per layer 4 flow
- | per refresh period



2. State information required inside each router

- | Identification of layer 4 flow (IP dest, Protocol, Port)
- | Identification of previous hop
- | Reservation state
- | Reserved resources

How to build services from these building blocks ?



Packet Marking

- What is the basic idea behind the two-bits architecture proposed in RFC2638 ?

How to solve integrated services scalability problems ?

- Signalling
 - No signalling protocol to indicate requirements of layer4 flows
 - Quality of Service is only provided to a few large aggregated flows
 - Markings inside packets indicate the QoS expected for each packet
 - Only a small number of different QoS can be supported

How to solve integrated services scalability problems ?

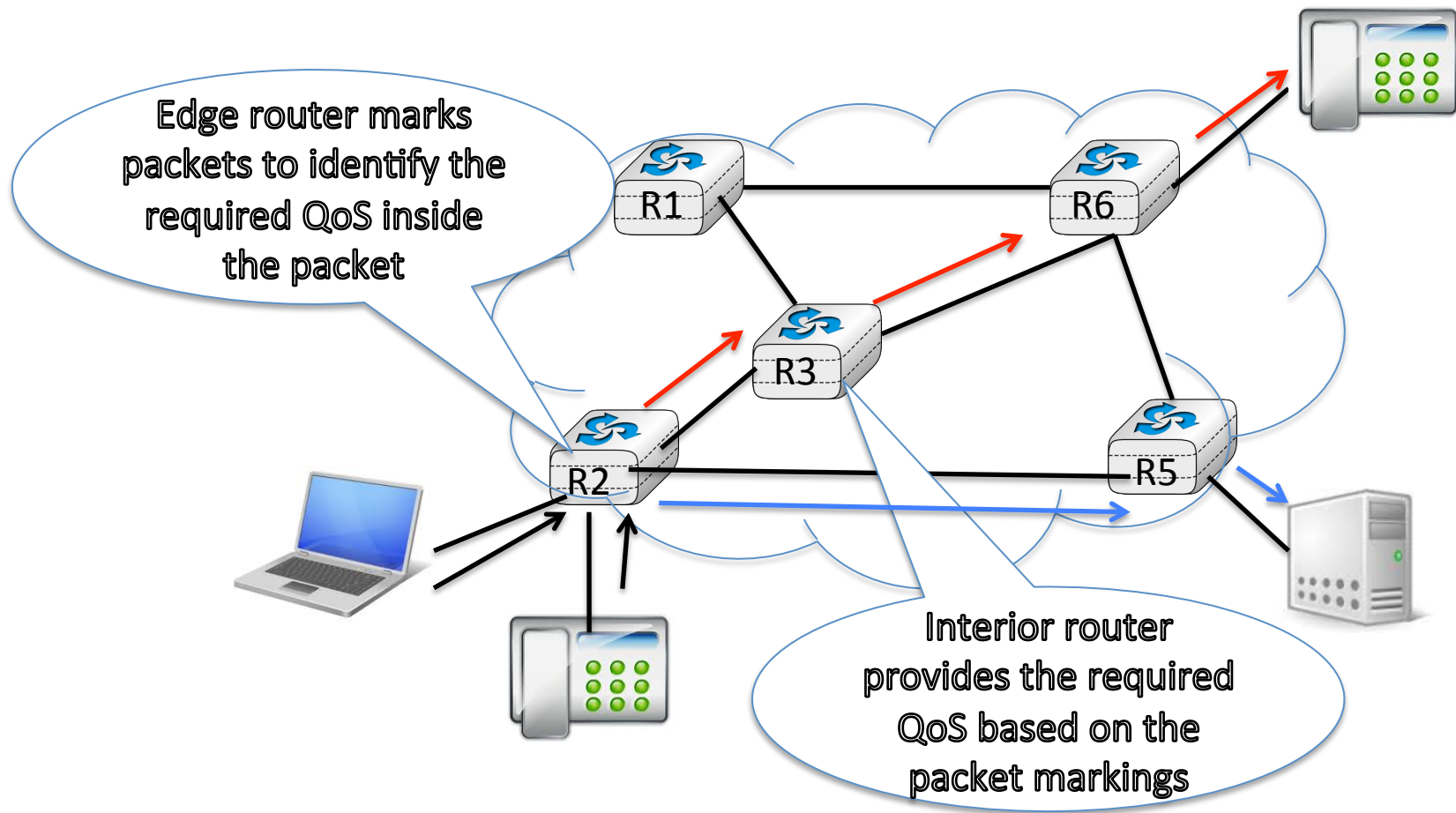
- State information inside the routers
 - No state maintained about end-to-end flows since there is no signalling protocol
 - Some configuration information on each router on a per service basis
 - Only a few services are supported inside the network

How to solve integrated services scalability problems ?

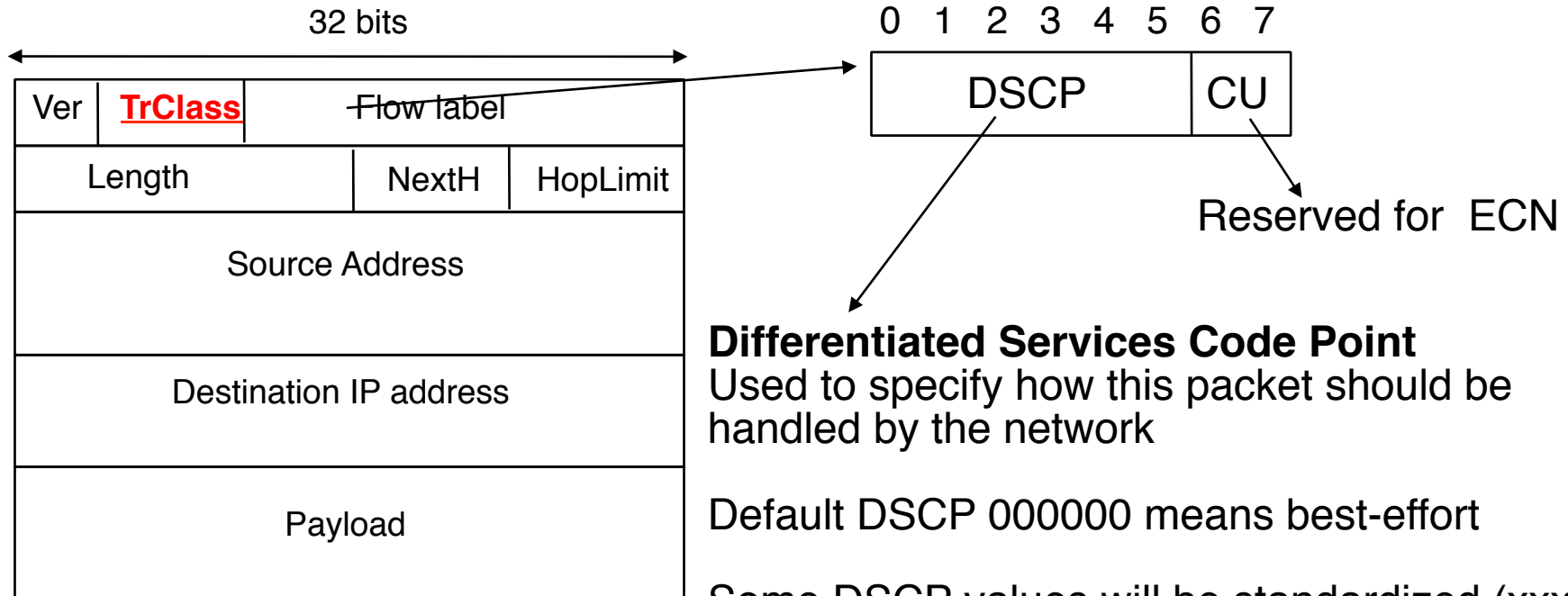
- Data plane
 - A few different services implies a small number of service dependant configuration on the traffic control mechanisms

Differentiated Services

- Basic architecture in an isolated network



How to mark IP packets ?

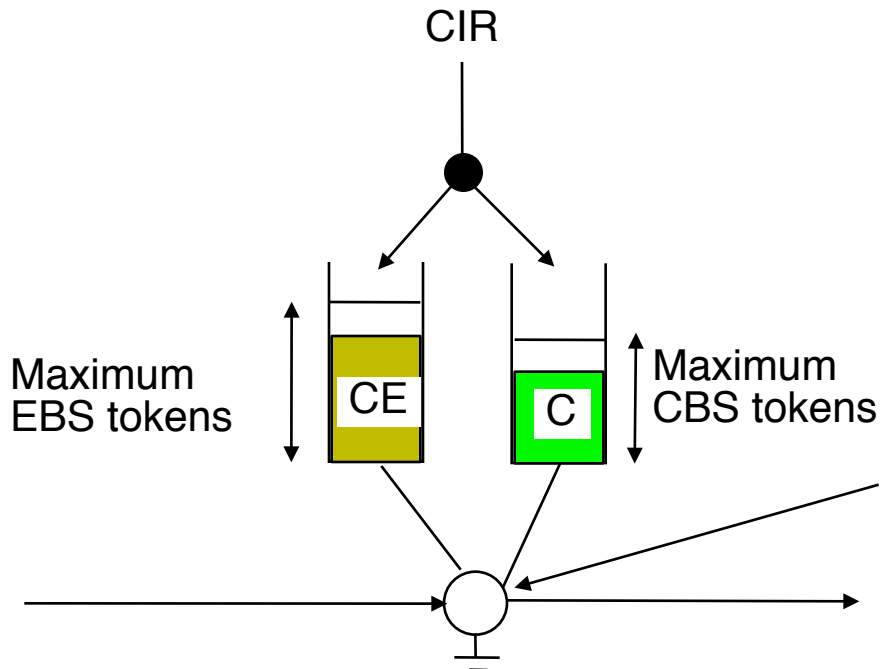


Packet marking techniques

- Policing and shaping
 - Performed only on the edge routers
 - Why ?

Single rate three color marker

- Committed Information Rate (CIR)
- Committed Burst Size (CBS)
- Excess Burst Size (EBS)



Bucket filling

Initialisation

```
C=CBS;  
CE=EBS;  
every 1/CIR second do  
{  
  if (C<CBS)  
  { C=C+1; }  
  else if (CE<EBS)  
  { CE=CE+1 }  
  else  
  { /* nothing */ }  
}
```

Arrival of packet P of length L

```
if (L <= C)  
{ /* packet is green */  
  C=C-L; }  
}  
else if (L <= CE)  
{ /* packet is yellow */  
  CE=CE-L; }  
else  
{ /* packet is red */ }
```

Two rate three colours marker

- Committed Information Rate (CIR)
- Committed Burst Size (CBS)
- Peak Information Rate (PIR)
- Peak Burst Size (EBS)

Peak Bucket filling

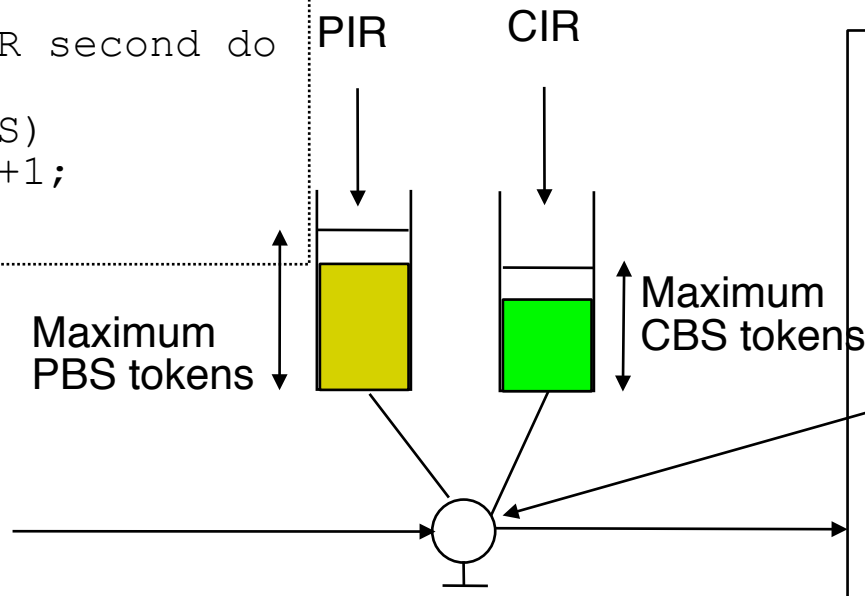
Initialisation

```
CP=PBS;  
every 1/PIR second do  
{  
  if (CP<PBS)  
    CP=CP+1;  
}
```

Committed Bucket filling

Initialisation

```
C=CBS;  
every 1/CIR second do  
{  
  if (C<CBS)  
    C=C+1;  
}
```



Arrival of packet P of length L

```
if (L > CP)  
{ /* packet is red */  
else if (L > C )  
  { /* packet is yellow */  
    CP=CP-L; }  
else  
  { /* packet is green */  
    CP=CP-L;  
    C=C-L;  
  }  
}
```

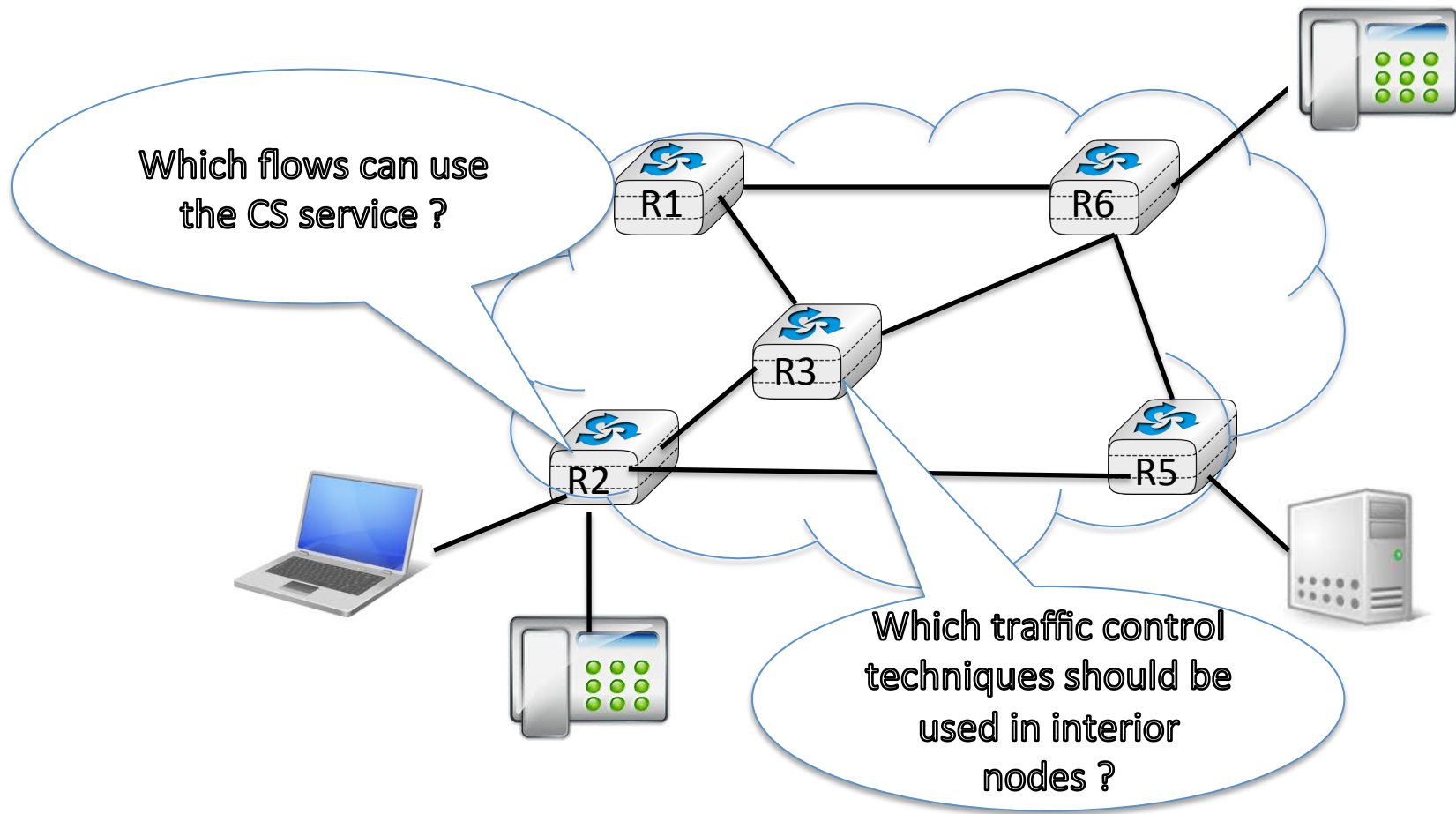
The Differentiated services

- Class Selector
- Assured Forwarding
- Expedited Forwarding
- Best Effort

Class Selector

- Definition
 - Use of class selector should yield to at least 2 independently forwarded traffic classes
 - Router should give a higher probability of timely forwarding to CS_x packets than CS_y packets if $x > y$
 - Packets with CS6 and CS7 should receive a better treatment than best-effort traffic
 - several routing protocols use CS6/7 IP packets

How to support Class Selector ?

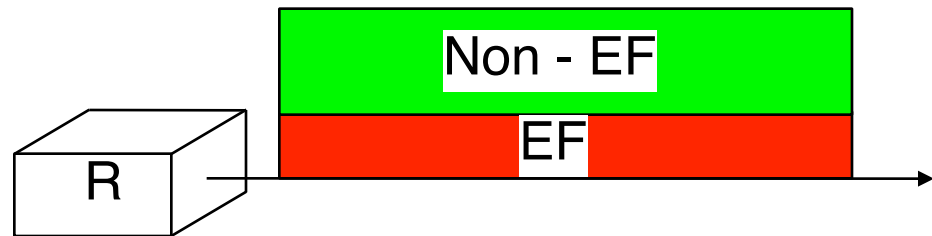


Class selector

- Which traffic control mechanisms should we configure on the edge routers ?
- Which traffic control mechanisms should we configure on the interior routers ?

Expedited Forwarding

- Objective
 - Provide low delay, low loss, low jitter service



- Principle
 - On each node, the administrator should be able to configure a rate R_{EF} for EF traffic
 - The EF traffic should be served at rate $\geq R_{EF}$ independently of the intensity of any other traffic through the node

Expedited Forwarding

- Which traffic control mechanisms should we configure on the edge routers ?
- Which traffic control mechanisms should we configure on the interior routers ?

Assured Forwarding

- Goal
 - provide differently provisioned services on top of a single IP network
 - business service with large over provisioning
 - premium ADSL service correctly provisioned
 - under provisioned cheap ADSL service

Assured Forwarding

- Principle
 - Color assigned to each traffic class
 - edge routers color packets based on their class
 - network resources are divided into a few classes
 - business service has a lot of resources to avoid congestion
 - premium ADSL has fewer resources, but can benefit from unused business resources
 - cheap ADSL service utilizes the leftover resources